Computation of Kendall's tau for Aids data.

Computation of Kendall's' tau for a given value of t consists of finding all proper rectangles which satisfy

$$r_i < t \leq z_i, \tag{1}$$

and all improper rectangles which satisfy

$$r_i < t \leq 23, \tag{2}$$

Suppose that $t = 10$. Then from Table 2 below we see that the proper observation rectangles 1-4, 10 and 13 satisfy condition (1) and so do their MI intersection rectangles

Table 2 Proper OR(first column), with its proper MI rectangle(2nd column) and in reduced form (third column) There are 27 proper ORs listed below.

Proper OR	Proper MI	Reduced (r,z)
$1(5,7) \times (11,12)$	$(5,7) \times (11,12)$	$(7,12)$
$2(1,10) \times (11,12)$	$(5,7) \times (11,12)$	$(7,12)$
$3(5,8) \times (13,13)$	$(5,7) \times (13,13)$	$(7,13)$
$4(1,7) \times (12,13)$	$(5,7) \times (13,13)$	$(7,13)$
$5(9,13) \times (14,15)$	$(10,11) \times (14,15)$	$(11,15)$
$6(8,10) \times (15,15)$	$(9,10) \times (15,15)$	$(10,15)$
$7(10,11) \times (14,15)$	$(10,11) \times (14,15)$	$(11,15)$
$8(10,12) \times (15,16)$	$(10,11) \times (15,16)$	$(11,16)$
$9(10,14) \times (15,16)$	$(10,11) \times (15,16)$	$(11,16)$
$10(1,7) \times (16,16)$	$(1,7) \times (16,16)$	$(7,16)$
$11(10,11) \times (15,16)$	$(10,11) \times (15,16)$	$(11,16)$
$12(10,12) \times (16,17)$	$(10,12) \times (16,17)$	$(12,17)$
$13(3,7) \times (17,17)$	$(3,7) \times (17,17)$	$(7,17)$
$14(9,13) \times (17,18)$	$(9,11) \times (17,18)$	$(11,18)$
$15(13,14) \times (17,18)$	$(13,14) \times (17,18)$	$(14,18)$
$16(13,15) \times (17,18)$	$(13,14) \times (17,18)$	$(14,18)$
$17(9,12) \times (17,18)$	$(9,11) \times (17,18)$	$(11,18)$
$18(13,15) \times (17,18)$	$(13,14) \times (17,18)$	$(14,18)$
$19(9,11) \times (17,18)$	$(9,11) \times (17,18)$	$(11,18)$
$20(7,9) \times (19,19)$	$(7,9) \times (19,19)$	$(9,19)$
$21(12,13) \times (20,20)$	$(12,13) \times (20,20)$	$(13,20)$
$22(13,14) \times (19,20)$	$(13,14) \times (19,20)$	$(14,20)$
$23(1,15) \times (20,21)$	$(13,14) \times (20,21)$	$(14,21)$
$24\ (13,14) \times (20,21)$	$(13,14) \times (20,21)$	$(14,21)$
$25(9,12) \times (22,22)$	$(9,12) \times (22,22)$	$(12,22)$
$26(10,12) \times (23,23)$	$(10,12) \times (23,23)$	$(12,23)$
$27(14,15) \times (23,23)$	$(14,15) \times (23,23)$	$(15,23)$

We next check improper rectangles in Table 5 and see that no observation rectangle satisfies condition (2). Moving on to Table 6

Table 6. Type 1 improper ORs which are not of the form $OR_m = (L_m, L_m+1] \times (23, \infty]$ with their probabilities and in reduced form.

Improper OR	Improper MI	multiplicity	reduced $(r, 23)$
$(1,7) \times (23,\infty)$	$(5,7) \times (23,\infty)$	1	$(7,23)$
$(5,7) \times (23,\infty)$	$(5,7) \times (23,\infty)$	2	$(7,23)$
$(7,9) \times (23,\infty)$	$(8,9) \times (23,\infty)$	1	$(9,23)$
$(12,14) \times (23,\infty)$	$(12,13) \times (23,\infty)$	2	$(13,23)$
$(13,15) \times (23,\infty)$	$(14,15) \times (23,\infty)$	3	$(15,23)$
total		9	

we see that the first four rectangles (counting the second rectangle twice) satisfy condition (2).

In summary, for $t = 10$ the following observations in their reduced form should be considered for computation of Kendall's tau. We will compute Kendall's tau based on the reduced proper and improper rectangles.

From Table 2

Proper MI	Reduced (r,z)
$(5,7) \times (11,12)$	$(7,12)$
$(5,7) \times (11,12)$	$(7,12)$
$(5,7) \times (13,13)$	$(7,13)$
$(5,7) \times (13,13)$	$(7,13)$
$(1,7) \times (16,16$	$(7,16)$
$(3,7) \times (17,17)$	$(7,17)$

and from Table 6

Improper MI	multiplicity	reduced $(r, 23)$
$(5,7) \times (23,\infty)$	1	$(7,23)$
$(5,7) \times (23,\infty)$	2	$(7,23)$
$(8,9) \times (23,\infty)$	1	$(9,23)$

In case of $t = 10$ there will be many ties because almost all reduced MIs have 7 for the time of $1 \rightarrow 2$ transition, so next we consider the case of $t = 12$.

For $t = 12$, from Table 2, we get the following 13 reduced MI rectangles (second column) with which we want to compute Kendall's tau.

$$\begin{array}{ll}
(5,7)\times(13,13) & (7,13) \\
(5,7)\times(13,13) & (7,13) \\
(10,11)\times(14,15) & (11,15) \\
(9,10)\times(15,15) & (10,15) \\
(10,11)\times(14,15) & (11,15) \\
(10,11)\times(15,16) & (11,16) \\
(10,11)\times(15,16) & (11,16) \\
(1,7)\times(16,16) & (7,16) \\
(10,11)\times(15,16) & (11,16) \\
(3,7)\times(17,17) & (7,17) \\
(9,11)\times(17,18) & (11,18) \\
(9,11)\times(17,18) & (11,18)
\end{array}$$

$$(7,9)\times(19,19) \quad (9,19).$$

Since there are 13 proper MI rectangles there will be

$$\binom{13}{2} = \frac{13(12)}{2} = 78$$

different pairs . For each pair we would have to decide if it is concordant, discordant or incomparable. For example pair $(7,13),(11,15)$ is concordant because $7 < 11$ and $13 < 15$. The pair $(11,15)$ and $(7,17)$ is discordant because $11 > 7$ but $15 < 17$. Finally a pair $(7,14)$ and $(7,16)$ is incomparable, because there is a tie between the first component of each pair.

Later we will include the reduced improper rectangles that satisfy (**??**).with t=12. From Table 5 there are 9 such rectangles

Improper OR=Improper MI	multiplicity	reduced MI
$(9,10)\times(23,\infty)$	2	$(10,23)$
$(10,11)\times(23,\infty)$	7	$(11,23)$

and from Table 6

$$\begin{array}{l}(5,7] \times (23,\infty)\\ (8,9] \times (23,\infty)\\ (9,10] \times (23,\infty)\\ (10,11] \times (23,\infty).\end{array}$$

there are 4 such reduced MI rectangles.